Dhruv Thippasandra

+1 858-705-7178 thippasa@usc.edu linkedin.com/in/dhruv-thippasandra github.com/Dhruvq

Education

University of Southern California Aug 2025 – Dec 2027
Master of Science in Electrical Engineering (Data Science and Machine Learning)

University of California, Riverside Sept 2022 – Jun 2025
Bachelor of Science in Electrical Engineering

- GPA: 3.77; Honors: Graduated **Cum Laude**; **Chancellor's Honors List** (all years); **Dean's Honors List** (all quarters)

Employment

Software Development Engineer Intern Sept 2026 – Present
AWS (Amazon Web Services) – Santa Clara, CA (Onsite)

- Designing a repeatable evaluation framework on **Amazon SageMaker AI** to benchmark **fine-tuned open-weight models** against a **frontier closed-weight model** with an inference-time harness (**RAG**, tools, memory, prompt/context engineering), so comparisons can be rerun on new tasks and model releases.
- Fine-tuning open-weight models and evaluating them against the harness baseline on **3 tasks** built on open-source datasets (e.g., **ABCD** customer-service dialogues), measuring **quality, cost and latency** under matched conditions.
- Producing task-level findings on where **fine-tuning** outperforms **frontier + harness** and where it does not, to inform internal guidance on when to recommend customization to customers.

Software Engineering Intern Jun 2026 – Aug 2026
AT&T – Dallas, TX (Onsite)

- Using **FastAPI** and **Python**, engineered an asynchronous backend anomaly detection service with comprehensive unit and integration test suites, deploying an **Extended Isolation Forest (EIF)** model to safely detect and suspend telecom fraud.
- Using **Azure**, **Snowflake**, and **PostgreSQL**, built automated ETL pipelines ingesting **12M+ phone records**, designing incremental upserts over full-table reloads to cut processing load and trigger bi-hourly batch fraud inference.
- Deployed a continuous production **MLOps pipeline** with **Docker** and **Azure**, retraining models on **5M+** monthly records with an estimated **sub-0.5% false-positive rate** to safely trigger automated suspensions.
- Using **Next.js** and **PostgreSQL**, implemented server-side data aggregation powering an internal trust and safety dashboard with multi-window audit views (daily to all-time) for hundreds of flagged fraud records.

Projects

Network Degradation Prediction — Python, XGBoost, Docker, AWS EC2, GitHub Actions CI/CD, Flask

- Engineered an end-to-end telemetry pipeline analyzing live RIPE Atlas Root DNS latency, training an **XGBoost** classifier with dynamic Z-score thresholding to predict degradation 5 minutes ahead with **91% accuracy**.
- Decoupled ML inference from live serving on **AWS EC2** using cron background workers and disk-cached history, ensuring **sub-100ms** dashboard response times without cold-start delays.

Wildfire Spread Forecaster — FastAPI, Python, Next.js, Maplibre GL, Proxmox, Nomad, Cloudflare

- Architected a 3-node **Proxmox** GPU cluster managed with **Nomad** to save ~**$400/month** in cloud compute costs, hosting the live wildfire forecaster over secure **Cloudflare Tunnels** with zero open inbound ports.
- Built an asynchronous **FastAPI** pipeline using Python **asyncio** to concurrently stream NASA, NOAA, and USGS satellite data, eliminating sequential I/O bottlenecks across geospatial endpoints.
- Implemented an empirically-driven Cellular Automata simulation engine modeling wind and slope vectors, computing 1–8 hour fire spread forecasts in **under 10 seconds** at **75.4% IoU accuracy**.

Apollo: Autonomous AI Research Analyst — Python, Gemini API, Gemma, SQLite, RAG, GitHub Actions

- Built an autonomous publication pipeline ingesting ~**1,500 arXiv preprints** biweekly, applying a 3-tier ranking engine (heuristic keywords, author tracking, and Gemma 3) to publish the **top 25** papers.
- Developed an interactive Telegram research bot backed by **Gemini 2.5** and a hybrid vector/keyword **SQLite** store, enabling real-time, grounded semantic Q&A with **under 5-second** retrieval latencies.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, HTML/CSS, Bash

Frameworks & Libraries: FastAPI, Next.js, React, Node.js, PyTorch, XGBoost, scikit-learn

Databases & Tools: PostgreSQL, Snowflake, SQLite, Git, GitHub Actions, Docker, Linux, CI/CD

Cloud & Infrastructure: AWS (SageMaker, EC2), Azure, Cloudflare, Proxmox, Nomad